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Awaworyi, Sefa; Yew, Siew Ling; Ugur, Mehmet

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# **Does government size affect per-capita income growth? A Hierarchical meta-regression analysis**

Sefa Awaworyi Churchill<sup>1</sup>, Siew Ling Yew<sup>2</sup>, Mehmet Ugur<sup>3</sup>

<sup>1,2</sup>Department of Economics, Monash University, VIC 3800, Australia

<sup>3</sup>Business School, University of Greenwich, London, UK

<sup>1</sup>Email: [sefa.awaworyi@monash.edu](mailto:sefa.awaworyi@monash.edu)

<sup>2</sup>Email: [siew.ling.yew@monash.edu](mailto:siew.ling.yew@monash.edu)

<sup>2</sup>Email: [m.ugur@gre.ac.uk](mailto:m.ugur@gre.ac.uk)

## **Abstract**

We conduct a hierarchical meta-regression analysis to review 87 empirical studies that report 769 estimates for the effects of government size on economic growth. We follow best-practice recommendations for meta-analysis of economics research, and address issues of publication selection bias and heterogeneity. When size is measured as the ratio of total government expenditures to GDP, the partial correlation between government size and per-capita GDP growth is negative in developed countries, but insignificant in developing countries. When size is measured as the ratio of consumption expenditures to GDP, the partial correlation is negative in both developed and developing countries, but the effect in developing countries is less adverse. We also report that government size is associated with less adverse effects when primary studies control for endogeneity and are published in journals and more recently, but it is associated with more adverse effects when primary studies use cross-section data. Our findings indicate that the relationship between government size and per-capita GDP growth is context-specific and likely to be biased due to endogeneity between the level of per-capita income and government expenditures.

**JEL code:** O40, H50, C1

**Keywords:** Economic growth; Government size; Government expenditure; Government consumption; Meta-analysis; Evidence synthesis

## **1. Introduction**

One of the most contentious issues in economics is whether 'big government' is good or bad for economic growth. From the mid-1980s onwards, the received wisdom has been that big government is detrimental to growth. This consensus has been tested by the onset of the recent financial crisis in Europe and the United States (US), which has called on governments to act not only as lenders of last resort but also as demand-managing and bank-nationalising fiscal heavyweights, the spending capacity of which was a crucial ingredient for recovery. Yet, it would be premature to claim that the consensus has lost its appeal. The financial crisis has also meant that a large number of European countries (including not only fiscally-vulnerable Greece, Ireland, Italy, Portugal and Spain, but also fiscally-comfortable countries such as Germany and the UK) had to embark on austerity programs with the aim of not only reducing budget deficits but also creating room for private investment as an engine of growth.

The continued appeal of the received wisdom may be due to ambiguity in the economic theory. Theoretically, big government can have both negative and positive effects on growth. On the one hand, big government can affect growth adversely because of crowding-out effects on private investment (Landau, 1983; Engen and Skinner, 1992). Big government also implies high taxes, most of which are growth-reducing due to their distortionary nature (De Gregorio, 1992). Increased government size can also be a source of inefficiency due to rent seeking and political corruption that harm economic growth (Hamilton, 2013; Gould and Amaro-Reyes, 1983; Mauro, 1995).

On the other hand, however, government can play a growth-enhancing role by providing public goods, minimising externalities, ensuring rule of law, and maintaining confidence in a reliable medium of exchange. Government also can contribute to growth by enhancing human capital through investments in health and education and by building and maintaining a sound infrastructure (Kormendi and Meguire, 1985; Ram, 1986; De Witte and Moesen, 2010). From a Keynesian perspective increased government spending increases aggregate demand that in turn induces an increase in aggregate supply.

The empirical literature on the issue is also inconclusive. On the one hand, a large number of studies report a negative relationship between government size and growth (see, e.g., Grier and Tullock, 1989; Barro, 1991; Lee, 1995; Ghura, 1995; Fölster and Henrekson, 2001; among others). On the other hand, a sizeable number report a positive relationship (see, e.g., Aschauer, 1989; Munnell, 1990; Evans and Karras, 1994).

Such heterogeneous findings are not surprising not only because of the ambiguity of the theoretical predictions but also because government size is usually measured differently,<sup>1</sup> countries may be at different stages of development, and they may have different political/institutional structures that affect the optimal government size (Bergh and Karlsson, 2010). In addition, model specification as well as estimation methods differ between studies. Beyond such sources of heterogeneity, findings from earlier studies have indicated that the effect of government size on growth is either not robust to model specification (Levine and Renelt, 1992) or does not remain a significant determinant of growth in a series of Bayesian averaging trials (Sala-i-Martin et al., 2004).<sup>2</sup>

Given this landscape, there is evident need to synthesize the existing evidence on the relationship between government size and growth. The synthesis should address not only the question of whether government size is growth-enhancing or growth-retarding, but also the sources of heterogeneity in the evidence base, and whether or not the reported evidence is robust to endogeneity and reverse causality, data type and model specification.

A number of past reviews have already tried to synthesize the existing findings (see, e.g., Poot, 2000; Nijkamp and Poot, 2004; and Bergh and Henrekson, 2011). Some of these reviews focus on the growth-effects of specific types of government expenditures; but others (e.g. Bergh and Henrekson, 2011) focus explicitly on government size and growth. These reviews provide useful narrative syntheses that reflect the state of the art in the

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<sup>1</sup> Government consumption tends to capture public outlays that are not productivity enhancing. It is often measured as all government expenditures for goods and services but excluding government military expenditure, and in some cases education expenditures (see, e.g., Barro, 1991; Easterly and Rebelo, 1993). According to the OECD, total government expenditure, on the other hand, captures the total amount of expenditure by government that needs to be financed via government revenues.

<sup>2</sup> In contrast, Sala-i-Martin et al. (2004) report that government consumption has a negative effect on growth (-0.034). Surprisingly, however, the adverse effect of government consumption is less severe than that of public investment (-0.062).

research field, but they are not suited to make inferences about either the 'effect size' or the effects of the moderating factors on the heterogeneity in the evidence base for three reasons. First, they do not take account of all effect-size estimates reported by the primary studies. Secondly, their selection of the primary studies is not systematic and as such their findings may be subject to sample selection bias. Last but not least, they do not address the risk of publication selection bias that arises when authors or editors are predisposed to publish significant findings or findings that support a given hypothesis.

Thus, this study aims to contribute to the existing literature in three areas. First, we address the issue of publication bias and establish whether government size has a 'genuine' growth effect beyond publication bias. Second, we examine the potential sources of heterogeneity in the evidence base and demonstrate how these moderating factors influence the reported effect-size estimates. Third, we synthesize the evidence with respect to: (a) two measures of government size (total government expenditures and government consumption); and (b) both developed and less developed countries (LDCs).

## 2. Theoretical and Empirical Considerations

Several perspectives exist on what explains growth.<sup>3</sup> The neoclassical growth model (Solow, 1956; Swan, 1956) has a successful track record in estimating capital and labour elasticities (i.e., capital and output shares in output) at the macro, industry and firm levels. However, the model has a number of shortcomings.<sup>4</sup> From the perspective of this review, the most important implication of the Solow-type growth model is that it has to be augmented with a measure of government size even though the latter is not theorised to have any effect on long-run growth rates.

In endogenous growth models, cross-country differences in per-capita income can persist indefinitely. These models have the advantage of taking into account the out-of-steady-state dynamics and the factors that affect the level of technology (Romer, 1986; Barro, 1990). In these models, government policy can alter the level of endogenous variables such as human capital or investment rates; and may have theory-driven implications for the country's long-run growth.

Hence, one factor that is likely to cause heterogeneity in the empirical evidence is the theoretical model utilised in the empirical studies. Some studies augment the neo-classical growth model with government size ( $G$ ), which yields a generic model of the type below:

$$Y_{it} = Ae^{\lambda t} K_{it}^{\alpha} L_{it}^{\beta} G_{it}^{\gamma} e^{u_{it}} \quad (1)$$

Dividing the inputs and the output by labour ( $L$ ), allowing for flexible returns to scale, taking natural logarithms and denoting  $\log(A)$  with  $\eta_i$ , we obtain:

$$y_{it} = \eta_i + \lambda t + \alpha k_{it} + \beta l_{it} + \gamma g_{it} + u_{it} \quad (\text{in levels})$$

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<sup>3</sup> See Poot (2000) and Bergh and Henrekson (2011) for a detailed overview on the theoretical arguments and models on the determinants of growth.

<sup>4</sup> First, the model is about steady-state level of income. Deviation of a country from its steady-state is temporary or random. Secondly, the main source of growth (i.e., technology) is exogenous, not observed in the model and the resulting total factor productivity (TFP) can be captured only through the residuals of the estimated model. Finally, the model predicts unconditional cross-country convergence of the per-capita income levels – a prediction that runs counter to evidence indicating persistent divergence in cross-country per-capita income levels.

(1a)

$$\Delta y_{it} = \Delta \lambda t + \alpha \Delta k_{it} + \beta \Delta l_{it} + \gamma \Delta g_{it} + v_{it} \quad (\text{in terms of growth rates})$$

(1b)

Specifications that include  $g_{it}$  instead of  $\Delta g_{it}$  as a regressor are widely used for assessing the impact of government size on economic growth, as the studies by Robinson (1977) and Landau (1983) indicate, thus the estimated growth model may take the form

$$\Delta y_{it} = \Delta \lambda t + \alpha \Delta k_{it} + \beta \Delta l_{it} + \gamma g_{it} + \varepsilon_{it}$$

(1c)

In (1a), subscripts  $i$  and  $t$  represent country and time, respectively.  $Y$  is income level, which is a function of capital stock ( $K$ ) and labour force ( $L$ ). Augmenting (1) with government size ( $G$ ) has been proposed by Feder (1983) and Ram (1986); and  $G$  can be measured as government expenditures or government consumption.<sup>5</sup> The small-case letters ( $k, l, g$ ) are the logarithms of capital per-capita, labour and government size. Finally,  $\eta_i$  is country fixed effect,  $\lambda t$  is time trend and  $u_{it}$  is an idiosyncratic error term.

Empirical studies included in this review, such as Grossman (1990), Atesoglu and Mueller (1990) and Dar and AmirKhalkhali (2002) among others, utilize version (1c) of the growth model. The parameter of interest,  $\gamma$ , is the effect of government size on per-capita income growth. All other variables are as defined in 1(a).

Government size ( $G$ ) is measured as total expenditures or consumption expenditures as a ratio of income (usually, gross domestic product – GDP). Some studies use the ratio of government size to GDP in level ( $G$ ), some others use the logarithm of the ratio ( $g$ ) providing an elasticity estimate of the effect size. In the former, the effect size ( $\gamma$ ) is not comparable across studies if the metrics with which  $G$  and  $Y$  are measured are different; in the latter, the effect size ( $\gamma$ ) is comparable across studies irrespective of the metrics with which  $G$  and  $Y$  are measured.

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<sup>5</sup> In (1a), constant returns to scale are not imposed. Hence the coefficient on labour ( $l$ ) indicates increasing, decreasing or constant returns to scale – depending on whether  $\beta$  is greater than, smaller than or equal to zero.

A considerable number of studies also adopt a variant of the endogenous growth model. These studies follow Barro (1991) and Mankiw et al. (1992), where the determinants of growth include investment in physical capital ( $K$ ) and human capital ( $H$ ), augmented with other covariates such as government expenditures ( $G$ ) and other variables found to be related to growth in the empirical literature (e.g., initial level of per-capita GDP, openness, financial development, etc.). The endogenous models usually take the following form:

$$\Delta y_{it} = \alpha K_{it} + \beta H_{it} + \gamma G_{it} + \sum_j \theta_j Z_{j,it} + \varepsilon_{it} \quad (2)$$

Here,  $\Delta y_{it}$  is the growth rate of per-capita GDP and  $G$  is the share of government expenditure to GDP.  $Z_j$  is a vector of variables commonly used in the economic growth literature, including financial deepening or institutional quality, etc. Subscripts  $i$  and  $t$  represent country and time, respectively. This specification has been widely used in the empirical growth literature, including Barro and Sala-i-Martin (1995), Stroup and Heckelman (2001) and Bose et al. (2007).

There are also hybrid models - including simultaneous equation models where government size is modelled to have both direct and indirect effects on growth and Keynesian models where government size affects growth from both the demand and supply side (see, e.g., Tanninen, 1999; Ghosh and Gregoriou, 2008). We include studies using hybrid models, but we consider only their estimates of the direct effect.

We meta-analyse the effect-size estimates only if they are reported by studies that utilize: (a) the ratio of government expenditures or consumption to GDP as the *independent variable*; and (b) the growth rate of per-capita GDP as the *outcome variable*.

To account for heterogeneity due to model choice, we control for endogenous growth models and treat the exogenous model as the reference category. Furthermore, both neoclassical and endogenous growth models (1c and 2) are used with panel and/or cross-section data averaged over a long period. Furthermore, panel-data studies may report estimates based on annual data or on data averaged over some years to smooth out the business cycle. These variations are additional sources of heterogeneity in the evidence



base. To account for this, we code each estimate not only with respect to the growth model it is derived from but also with respect to data type and the number of years over which the panel data is averaged.

Other sources of heterogeneity include: (i) whether primary studies control for endogeneity through instrumental variable or dynamic panel-data techniques; (ii) the range of control variables included in estimated models; (iii) publication type (e.g., journal articles, working papers, book chapters, reports, etc.); (iv) publication date; (v) journal quality ranking; (vi) country type (developed *versus* less developed); (vii) length of periods over which data is averaged in cross-section and panel-data studies; and (viii) the data period. We code the effect-size estimates accordingly and control for these sources of heterogeneity within a multiple meta-regression setting.

### **3. Data and an overview of the evidence base**

Our meta-analysis methodology draws on guidelines proposed by the Meta-analysis of Economics Research Network (MAER-Net), which reflects best practices and transparency in meta-analyses (Stanley et al., 2013). We searched five electronic databases - JSTOR, EconLit, Business Source Complete, Google Scholar and ProQuest - for journal articles, working papers and book chapters; using various keywords for government size and growth.<sup>6</sup> We also conduct a manual search where we examine the references of key reviews and studies that examine the government size-growth relationship to ensure that other relevant studies are included in our meta-analysis.

In this study, we include all effect-size estimates reported by empirical studies that examine the direct effect of government size on growth. The independent variable must be either total government expenditure or government consumption expenditure, and must be measured as a proportion or share of GDP. The dependent variable must be the growth rate of per capita GDP. Thus, we exclude studies that use other measures of government size such as expenditure or consumption levels or the growth rate of government

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<sup>6</sup> The keywords for government size include total government expenditures, government consumption, government spending, outlays, public spending, public expenditures, and public consumption. Keywords for economic growth include GDP, per capita income, growth, economic performance, and economic activity.

expenditures/consumption. We also exclude studies that estimate the effects of government size on GDP levels instead of per-capita GDP growth.

Adhering to the above inclusion/exclusion criteria, our meta-analysis consists of 87 primary studies reporting 769 estimates on the relationship between government size and growth. Table 1A presents an overview of the studies that report on the effect of total government expenditure, while Table 1B presents the overview of those reporting on the effect of government consumption only.

#### **4. Meta-analysis tools and findings**

The meta-analysis methods adopted in this study have been used previously in various studies (see, e.g., Abreu et al., 2005; Alptekin and Levine, 2012; Stanley and Doucouliagos, 2012 and Ugur, 2014). Therefore, we elaborate on the methods and tools in the Appendix and present the results in the main text to achieve brevity and continuous flow.

##### **4.1. Partial correlation coefficients (PCCs) and fixed effect weighted means (FEWMs)**

Given the variety of the metrics with which the dependent and outcome variables are measured, we calculate PCCs for each effect-size estimate to allow for comparability between and within primary studies (see Appendix). We also calculate FEWMs of the PCCs for each primary study and for two samples consisting of studies that investigate the growth effects of total government expenditures (Table 1A) and government consumption (Table 1B). As summary measures, FEWMs are considered more reliable than simple means, and also less affected by publication bias compared to random effects weighted averages (Stanley, 2008; Henmi and Copas, 2010; Stanley and Doucouliagos, 2014).

Tables 1A and 1B Here

Of the 53 primary studies that investigate the relationship between total government expenditures and per-capita GDP growth (Table 1A), 17 studies (32.08% of the total) report 84 estimates (20.44% of the total) that are insignificant; 26 studies (49.06%) report 201 estimates (48.90%) that are negative and significant; and 10 studies (18.87%) report

126 estimates (30.66%) that are positive and significant. The overall FEWM for all 411 estimates is -0.0083, indicating a negative association between total government expenditure and growth. The effect size is however statistically insignificant. Furthermore, this effect-size estimate, according to guidelines proposed by Cohen (1988) and Doucouliagos (2011), is too small to be practically significant – either as a measure of effect size or as a guide for policy.

In Table 1B, 46 primary studies with 388 estimates examine the relationship between government consumption and economic growth. The results indicate that the FEWMs are statistically insignificant in 14 primary studies (30.43% of total studies) that report 82 estimates (21.13% of the total). Of the remaining 32 studies, 31 (67.39% of total studies) with 290 estimates (74.75% of total estimates) yield negative and significant FEWMs, while the remaining one study with 16 estimates (4.12% of total estimates) yields a positive and significant FEWM. The overall FEWM for all 388 estimates is found to be -0.1204, indicating a negative association between government consumption and growth. According to guidelines indicated above, this represents a small effect with practical significance.

FEWMs are valid only if the effect-size estimates reported by primary studies are not subject to selection bias. To test whether selection bias exists in the evidence base and obtain average effect-size estimates corrected for selection bias, we conduct funnel-asymmetry and precision-effect tests (FAT/PET) in the next section.

#### **4.2. Effect size beyond publication selection**

The precision-effect test (PET) and the funnel-asymmetry test (FAT) involve the estimation of a weighted least square bivariate model, in which the effect-size estimate is a function of its standard error (see Egger et al., 1997; Stanley, 2008). Furthermore, Stanley and Doucouliagos (2007) propose that precision-effect estimation with standard errors (PEESE) can be used to obtain an estimate corrected for non-linear relationship between effect size and its standard error (see Appendix). The PEESE should be conducted only if PET/FAT procedure indicates the existence of a genuine effect beyond publication selection bias.

We estimate PET-FAT-PEESE models for two measures of government size: total government expenditures and government consumption expenditures; and for developed and LDCs. Our estimates are obtained using a hierarchical linear model (HLM) specification (Goldstein, 1995), whereby individual effect-size estimates are nested within studies reporting them. The choice of HLM is informed by the data structure at hand. The effect-size estimates are nested within each study where data, model specification and estimation method are significant sources of dependence between reported estimates. The appropriateness of the HLM is verified through a likelihood ratio (LR) test that compares the HLM with standard OLS; whereas the type of HLM is determined by LR tests that compare random effects intercepts with random effects intercepts and slopes.<sup>7</sup> The PET-FAT-PEESE results are presented in Tables 2A and 2B, for two full samples and for two country types (developed and LDCs) within each sample.

#### Tables 2A and 2B Here

Regarding total government expenditure and growth, we find no evidence of genuine effect in the full sample or in LDCs as the coefficient of the precision is statistically insignificant (columns 1 and 3 of Table 2A). In the developed countries sample (column 2 of Table 2A), we find evidence of a negative effect (-0.13) without evidence of publication selection bias. This PET-FAT result is also supported by the PEESE result (Column 4) with a slightly more adverse effect (-0.14). Thus, with respect to total government expenditure as a proxy for government size, we report a negative partial correlation with growth in developed countries only.

With regard to government consumption and growth (Table 2B), we find evidence of a negative effect together with significant negative publication selection bias for the entire sample (column 1) and for the developed countries sample (column 2), but no significant effect for the LDC sample (column 3). PEESE results that correct for non-linear relations between effect-size estimates and their standard errors (columns 4 and 5) confirm the

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<sup>7</sup> The HLM is employed to deal with data dependence by De Dominicis et al. (2008), Bateman and Jones (2003), and Alptekin and Levine (2012), among others. The likelihood ratio test results that compare HLM with OLS and the types of HLM structures are available on request.

existence of negative effects for the full sample and for developed-country sample (-0.10 and -0.14, respectively).

Statistical significance in the empirical literature has been clearly distinguished from economic (or practical) significance, especially when the size of a statistically significant coefficient is small (Ziliak and McCloskey, 2004). Cohen (1988) indicates that an estimate represents a small effect if its absolute value is around 0.10, a medium effect if it is 0.25 and over, and a large effect if it is greater than 0.4. Doucouliagos (2011) argues that the guidelines presented by Cohen (1988) understate the economic significance of empirical effect when partial correlation coefficients (PCCs) are used. Thus, Doucouliagos (2011) suggests that PCCs larger than the absolute value of 0.33 can be considered as indicators of 'large' effect, while those above (below) 0.07 can be considered 'medium' ('small').

In the light of the guidance proposed by Doucouliagos (2011), these findings indicate that: (i) total government expenditures have a medium and adverse effect on per-capita income growth in developed countries only; (ii) the effect of government consumption on per-capita income growth is also medium in developed countries and in developed and LDCs pooled together; and (iii) neither total expenditures nor government consumption has a significant effect on per-capita income growth in LDCs.

However, both FEWMs and PEESE results have limited applicability because they conceal a high degree of variation in the evidence base. Indeed, the coefficient of variation for full-sample PCCs in Table 1A and 1B are 9.11 and 1.28, respectively. In addition, the FEWMs and PEESE results are based on the assumption that, apart from the standard errors, all other moderating factors that affect the reported estimates are either zero (in the case of FEWMs) or at their sample means (in the case of PEESE). This assumption is too restrictive because the moderating factors that influence the effect-size estimates reported in primary studies differ between studies and between estimates reported by the same study. Therefore, it is necessary to identify the moderating factors (i.e., the sources of variation) in the evidence base and quantify their influence on the effect-size estimates reported in primary studies. This is done in the next section, followed by a detailed discussion of the

implications for the government size – growth relationship in the conclusions and discussion section.

### **4.3. Addressing Heterogeneity**

To identify the sources of heterogeneity and quantify their influence on the reported effect-size estimates, we estimate a multivariate meta-regression model (MRM) for each sample (i.e., for total government expenditures and government consumption). As indicated in the appendix, we estimate a general and a specific MRM for each sample. The general specification includes all moderating factors that can be measured on the basis of the information we obtain from the primary studies. However, the inclusion of all observable moderating factors poses issues of over-determination and multicollinearity. Therefore, we follow a general-to-specific model routine, which involves the exclusion of the moderating variables with high p-values (highly insignificant variables) one at a time until all remaining variables are statistically significant.<sup>8</sup>

Three main dimensions of primary study characteristics are captured by our moderator variables. These moderator variables are informed by the theoretical, empirical and methodological dimensions of primary studies, as well as other differences presented by primary studies that are likely to affect the government size-growth relationship. The first set of moderator variables capture the variations in econometric specifications and theoretical models adopted by the primary studies. The second category captures data characteristics in primary studies, and the third examines the publication characteristics of the primary studies. Summary statistics for moderator variables and their description are presented in Tables 3A and 3B.

Tables 3A and 3B Here

Results from the general and specific MRMs are presented in Table 4A (for total government expenditures) and Table 4B (for government consumption) below. The

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<sup>8</sup> See Campos et al. (2005) for a detailed discussion on the general-to-specific modelling literature.

general models include 20 moderating variables that capture the three dimensions of the research field indicated above. The paragraphs below summarize the findings and interpret their implications for the relationship between government size and per-capita income growth.

### ***Theoretical Models and Econometric Specifications***

To investigate if differences in underlying theoretical models affect the government size-growth relationship, we use a dummy variable that controls for studies that base their specification on endogenous growth models and exclude studies that use the Solow-type growth model as base. From Table 4A, we find that the underlying theoretical model does not have a significant effect on reported effect-size estimates when the latter are about the effects of total government expenditures on per-capita income growth. However, we note from Table 4B that studies that utilize an endogenous growth model tend to report more adverse effect-size estimates for the relationship between government consumption and per-capita income growth.

Tables 4A and 4B Here

With respect to econometric dimension, we first examine the difference between estimates based on cross-section data as opposed to panel-data or time-series data. This control is relevant because cross-section estimations overlook fixed effects that may reflect country-specific difference in preferences and technology. In the presence of fixed-effects, estimates based on cross-section data may yield biased results. Panel-data estimations can address this source of bias by purging the country-specific fixed effects and focusing on temporal variations in the data.

In Table 4A where the focus is on total government expenditures, we find that the use of cross-section data (as opposed to panel data) is associated with more adverse effects on growth, but the effect is insignificant. However, when government size is proxied by consumption (Table 4B), the use of cross-section data is associated with more adverse effects; and the coefficient is significant. We also control for studies that use time-series data as opposed to cross-section and panel data. In both total expenditures (Table 4A) and

consumption expenditures (Table 4B) samples, we find no statistically significant effects. Hence, we conclude that the use of cross-section data is likely to be a source of negative bias in the evidence base. Therefore, it is likely that the negative association between government consumption and growth referred to in the public and policy debate should be qualified.

The second dimension of the econometric specification we consider is model specification. In the empirical growth literature, it is well known that the inclusion (or exclusion) of certain regressors in growth regressions can affect the reported effect-size estimate. Levine and Renelt (1992) indicate that initial GDP per capita, investment share of GDP and population are important growth determinants. Hence, we include dummies for studies that control for these variables. We also include a dummy for studies that control for tax in their growth regressions, given that the distortionary effects of taxation feature as a major factor in the debate on government size and growth.

MRM results in Tables 4A and 4B confirm that the inclusion of these variables in growth regressions tend to affect the estimates reported in primary studies. For instance, in Table 4A, we find that studies that control for investment, population or initial GDP (compared to those that do not) tend to report more adverse effects. Results from Table 4B show that studies that control for investment (compared to those that do not) tend to report less adverse effects whereas the opposite is observed for those that control for population.

Therefore, the inclusion or exclusion of key determinants of growth in the regression significantly alters the magnitude of the reported effect size estimates, but the effect is not consistent across the measures of government size. Despite the variation, we argue that it would be good practice for researchers to include the key regressors in their regressions with a view to minimize the risk of model specification bias and the additional heterogeneity that would result from such biases.

Another dimension of the econometric specification that may affect the reported estimates concerns the length of time over which both regressors and regressands are averaged. Two arguments can be put forward in favour of averaging. First, averaging over a period equal



to the business cycle (usually five years) eliminates the effect of business cycle and this is particularly important if measures of business cycle (e.g., output gap) are not included in the model. Secondly, estimates based on data averaged over 5 years or more can be interpreted as medium- to long-run effects as opposed to short-run effects. Thus, to verify if estimates reported in primary studies are affected by the period of data averaging, we control for studies where data is averaged over five years or more, with others where annual data is used or the data is averaged over periods of less than 5 years. In both government consumption and total expenditure samples, we find that the data averaging period has no statistically significant effect on estimates reported in primary studies.

We further examine the nature of reported estimates for studies that use panel data and adopt data averaging of 5 years or more, and also those that use cross-section data with data averaging of 5 years and above (as opposed to those that do not). In the total expenditure specification, the coefficient for studies that use panel data with data averaging of 5 years or more is statistically insignificant. However, we find that studies that use cross-section data with data averaging of 5 years and above tend to report less adverse effects of government consumption on growth.

This finding indicates that the bias that results from failure to account for country fixed-effects in cross-section data is larger when the data averaging period is short. This is to be expected because country fixed effects are more likely to remain fixed over shorter time horizons. Another implication of this finding is that the relatively larger adverse effects reported by studies using cross-section data are likely to be driven by the dominance of the effect-size estimates based on short time horizons.

The last dimension relating to econometric and theoretical specification concerns the econometric methodology used by primary studies. In the empirical growth literature, various econometric methods have been used and these methodologies aim at addressing specific issues. For instance, OLS estimates have been found to be inconsistent and biased in the presence of endogeneity. To address endogeneity, some primary studies tend to use instrumental variable (IV) techniques. Therefore, we control for studies that control for endogeneity as opposed to those that do not. The coefficient on the dummy for studies that

control for endogeneity is found to be positive and significant in the total expenditure-growth specification but insignificant in the government consumption-growth specification. This suggests that studies that control for endogeneity (compared to those that do not) tend to report less adverse effects of total expenditure on growth.

### ***Data Characteristics***

With regards to data characteristics, we first examine if the government size-growth relationship is time variant. Thus, we include dummy variables to capture the ‘recentness’ of data and how data time periods affect reported estimates. We include dummy variables to capture the decade in which the beginning year of the data period falls, and the subsequent years in the data. For instance, a primary study that uses data from 1983 to 1999 will fall under “Data Period (1980+)”. Put differently, Data Period (1980+) captures studies where data years equal to or greater than 1980. Hence, we introduce dummies for data period 1960+, 1970+, 1980+, and 2000+ and exclude the 1950+ as base. MRA results for the total expenditure specification mainly show statistically insignificant coefficients for data period dummies. However, from Table 4B coefficients for data period dummies are negative and statistically significant. Particularly, we note that the magnitude of the coefficient increases as the decades increase. Thus, the smallest coefficient is observed for “Data Period 1960+” and the largest for “Data Period 1990+”. This suggests that studies that use newer datasets tend to report more adverse effects of government consumption on growth. This is also the case for total government expenditure as the only included dummy in the general-to-specific model (Data Period 2000+), is negative and statistically significant as well.

We also examine the effect of country type using a country-type dummy. Although PET-FAT results reveal a negative effect of big government on growth in developed countries, it is worthwhile to control for this in our MRA as well. This ensures the inclusion of all relevant moderator variables that capture the necessary dimensions. We therefore control for studies that report estimates using data on developed countries as opposed to those that use data from LDCs and a mixture of both developed and LDCs. Results from Table 4A confirm what the PET-FAT results suggest. The developed country dummy is significant

and negative, suggesting that studies that use data on developed countries tend to report more adverse effects of total expenditure on growth compared to those that use LDCs sample and mixed sample. From Table 4B, the coefficient of studies that use developed countries data is also negative but insignificant.

### ***Publication Characteristics***

Under the publication characteristics dimension, we first control for publication type. Here, we examine if journal articles tend to systematically report different effect sizes in comparison to book chapters and working papers. This allows us to determine whether researchers, authors and editors are predisposed to publishing and/or accepting studies with statistically significant results that are consistent with theory to justify model selection. Using book chapters and working papers as base, we include a dummy for journal articles in our MRA specification. Results reveal that studies published in journals tend to report less adverse effects of government size on growth. This is consistent across both measures of government size and specification type (i.e., both general and general-to-specific).

Furthermore, we examine if the publication outlet used by primary studies presents some variations in reported estimate. We therefore control for high-ranked journal as opposed to low-ranked journals.<sup>9</sup> From Table 4A, the coefficient for studies published in high-ranked journals is statistically insignificant. However, studies published in high-ranked journals tend to report more adverse effects of government consumption on growth (Table 4B).

Next, we control for publication year. Examining publication year enables us to identify whether more recent studies, as opposed to older studies, tend to report different estimates. Thus, we include dummy variables similar to those constructed for data period. For instance, studies published between 1998 and 2013 fall under “Publication Year (1990+)” and those published between 2001 and 2014 fall under “Publication Year (2000+)”. Leaving 1980+ as base, we control for studies published in the decades starting

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<sup>9</sup> The Australian Business Dean’s Council (ABDC) and the Australian Research Council (ARC) present classifications for journal quality. Journals are ranked in descending order of quality as A\*, A, B and C. Thus, we introduce a dummy for A\* and A ranked journals (high quality) in our MRA, and use other ranks as base.

1990, 2000, and 2010. In both government consumption and total government expenditure specifications, publication year dummies are significant. Specifically, from Table 4A, we note that dummies for 1990+ and 2000+ are negative whereas that of 2010+ is positive. Furthermore, the magnitude of the coefficient for studies published in the 2000s is smaller than those published in the 1990s. Thus, overall, less adverse effects are associated with newer studies that examine the effect of total expenditure on growth. For the government consumption specification, we observe that dummies for 1990+ and 2000+ are positive while that for 2010+ is negative.

## **5. Discussion and conclusions**

This paper reviews the empirical literature on the association between government size and economic growth. We focus on total government expenditures and government consumption expenditures (as a share of GDP) as measures of government size. Results are based on a synthesis of 87 studies solely examining the effect of the government size measures as specified above on per-capita GDP growth. We control for publication selection bias and address issues of heterogeneity in the existing literature.

Drawing on guidelines proposed by Doucouliagos (2011), the PET-FAT-PEESE results reported above indicate that the average effect of government size on growth, using both proxies of government size, is medium only in developed countries. The average effect of total government expenditures is insignificant in LDCs and when both developed and LDCs are pooled together. On the other hand, the average effect of government consumption is insignificant in LDCs, but it is medium in developed countries and when both country types are pooled together.

These findings suggest that the existing evidence does not support an overall inference that establishes a negative relationship between government size and per-capita income growth for several reasons including: (i) potential biases induced by reverse causality between government size and per-capita income; (ii) lack of control for country fixed effects in cross-section studies; and (iii) absence of control for non-linear relationships between government size and per-capita GDP growth.

Furthermore, the effect is specific to the level of development: government size tends to have a negative effect on per-capita income growth as the level of income increases. This finding ties in with the Armey curve hypothesis (Armey, 1995), which posits an inverted-U relationship between government size and economic growth. The theoretical argument here is that government size may be characterised by decreasing returns. Another theoretical argument relates to the distortionary nature of taxes, which is minimal for low levels of taxation, but beyond a certain threshold, they grow rapidly and become extremely large (Agell, 1996). Hence, our findings suggest that estimates of the relationship between government size and growth obtained from linear estimations may be biased (see also, Barro, 1990).

In addition, developed countries tend to have well-developed systems of automatic stabilisers such as social security expenditures and progressive taxation. According to the World Social Security Report 2010/11, Europe spends between 20 and 30 per cent of GDP on social security, while in most African countries social security spending accounts only for 4–6 per cent of GDP.

According to Devarajan et al. (1996), social security expenditures are unproductive and as such they may be driving the negative relationship between government size and per-capita income growth in developed countries. However, social security expenditures and other forms of automatic stabilisers may be conducive to lower growth rates because of the reverse causality they inject into the government size-growth relationship. As indicated by Bergh and Henrekson (2011), automatic stabilisers on the expenditure sides would increase as GDP falls. This well-known feature of the automatic stabilisers introduces a negative bias in the estimates for the effect of government size on growth. The risk of such bias is higher in developed countries with higher incidence of automatic stabilisers. At a more general level of observation, the risk of reverse causality is confirmed by our finding that studies that control for endogeneity tend to report less adverse effects of government size on growth.

Furthermore, the more pronounced negative effects for developed countries may be related to Wagner's Law, which indicates that government size increases with the level of

income. There is evidence indicating that the long-run elasticity of government size with respect to growth in developed countries is large (Lamartina and Zaghini, 2011). In this case, the government size-GDP ratio for developed countries will grow faster than LDCs for a given increase in GDP. This additional endogeneity problem leads to what Roodman (2008) describes as ‘the looking glass problem’. Roodman (1998) demonstrates that the aid-GDP ratio decreases as GDP increases and this endogeneity leads to strong positive effects of aid on GDP growth. This finding is opposite to what we observe in this study but proves the point that if government size-GDP ratio is endogenous to GDP (i.e., if Wagner’s Law holds), then the stronger negative effects reported on developed countries may be due to either lack of control for endogeneity in the growth regressions or absence of adequate instruments or both. This conclusion also ties in with our findings from the MRM (Table 4A) that studies that control for endogeneity report less adverse effects of government size on growth.

We also find that studies published in journals tend to report less adverse effects compared to working papers and book chapters. This is consistent across both measures of government size, and thus raises the question as to whether the negative association between government size and per-capita income growth may be driven by less rigorous external reviewing processes in the case of book chapters and working papers. However, we do not wish to overemphasize this because in the government consumption sample, we find that studies published in higher-ranked journals tend to report more adverse effects of government size on growth. This is an indication of the ‘Winner’s curse’ - whereby journals with good reputation capitalize on their reputation and publish ‘more selected’ findings (see Costa-Font et al., 2013; Ugur, 2014).

In conclusion, our findings show that where an evidence base is too diverse, meta-analysis can be highly effective in synthesizing the evidence base and accounting for the sources of heterogeneity among reported findings. Our findings in this study indicate that government size is more likely to be associated with negative effects on per-capita income growth in developed countries. They also indicate that the medium-sized adverse effects in developed countries may be biased due to endogeneity and reverse causality problems, which are

either unaddressed in a large segment of the evidence base or the instruments used to address these problems are weak or both. Therefore, we call for caution in establishing casual links between government size and per-capita income growth. We also call for use of non-linear models in the estimation of the government size – growth relationship. As indicated by Agell (1996), non-linear models may provide richer evidence on the optimal government size, particularly when the latter is measured in terms of tax revenues. Finally, and as indicated by Kneller et al. (1999), Poot (2000) and Bergh and Henrekson (2011), we call for further research on the relationship between particular components of the government size and growth as such studies are more likely to produce policy-relevant findings compared to studies that focus on total measures of government size.

## 6. References

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**Table 1A Means of the partial correlation coefficients per study: Total government expenditures and growth**

| Study                                | Reported estimates | Simple mean | *Coefficient of variation (simple mean) | Fixed-effect weighted mean (FEWM) | *Coefficient of variation (FEWM) | Significant | Confidence interval |
|--------------------------------------|--------------------|-------------|-----------------------------------------|-----------------------------------|----------------------------------|-------------|---------------------|
| Adam and Bevan (2005)                | 9                  | 0.2255      | 0.1130                                  | 0.2256                            | 0.1128                           | Yes         | (0.2060, 0.2452)    |
| Afonso and Furceri (2010)            | 10                 | -0.2955     | 0.2166                                  | -0.3092                           | 0.2295                           | Yes         | (-0.3600, -0.2585)  |
| Afonso and Jalles (2014)             | 12                 | -0.1718     | 0.5418                                  | -0.1651                           | 0.5900                           | Yes         | (-0.2270, -0.1032)  |
| Afonso and Jalles (2013)             | 13                 | -0.1491     | 0.6798                                  | -0.1292                           | 0.5945                           | Yes         | (-0.1756, -0.0828)  |
| Afonso et al. (2010)                 | 32                 | 0.0434      | 0.4580                                  | 0.0411                            | 0.3841                           | Yes         | (0.0354, 0.0468)    |
| Agell et al. (1997)                  | 3                  | -0.0723     | 2.0667                                  | -0.0828                           | 1.7525                           | No          | (-0.4430, 0.2775)   |
| Angelopoulos et al. (2007)           | 2                  | -0.2788     | 0.3596                                  | -0.2819                           | 0.3554                           | No          | (-1.1819, 0.6181)   |
| Angelopoulos et al. (2008)           | 18                 | -0.2056     | 0.9474                                  | -0.2245                           | 1.0092                           | Yes         | (-0.3372, -0.1119)  |
| Arin (2004)                          | 20                 | -0.3079     | 0.3402                                  | -0.2822                           | 0.3048                           | Yes         | (-0.3224, -0.2419)  |
| Bergh and Karlsson (2010)            | 9                  | -0.2705     | 0.3079                                  | -0.2652                           | 0.3208                           | Yes         | (-0.3306, -0.1998)  |
| Bergh and Öhrn (2011)                | 9                  | 0.0235      | 7.9725                                  | 0.0082                            | 2.6514                           | No          | (-0.1338, 0.1501)   |
| Bernhard (2001)                      | 2                  | -0.3821     | 0.1846                                  | -0.3843                           | 0.1833                           | No          | (-1.0173, 0.2486)   |
| Bojanic (2013)                       | 14                 | 0.3154      | 0.6995                                  | 0.3319                            | 0.6608                           | Yes         | (0.2053, 0.4586)    |
| Bose et al. (2007)                   | 2                  | 0.4337      | 0.0182                                  | 0.4339                            | 0.0182                           | Yes         | (0.3630, 0.5047)    |
| Butkiewicz and Yanikkaya (2011)      | 30                 | -0.1489     | 0.8045                                  | -0.1341                           | 0.8184                           | Yes         | (-0.1750, -0.0931)  |
| Chen and Lee (2005)                  | 9                  | 0.0470      | 9.3072                                  | -0.0975                           | 3.7851                           | No          | (-0.3812, 0.1862)   |
| Colombier (2009)                     | 4                  | 0.1617      | 3.6786                                  | 0.1499                            | 1.7667                           | No          | (-0.2715, 0.5712)   |
| Cooray (2009)                        | 10                 | 0.1094      | 0.2122                                  | 0.1095                            | 0.2122                           | Yes         | (0.0929, 0.1262)    |
| Dalic (2013)                         | 4                  | -0.2719     | 0.0920                                  | -0.2711                           | 0.0932                           | Yes         | (-0.3113, -0.2309)  |
| Dar and AmirKhalkhali (2002)         | 3                  | -0.1495     | 0.8056                                  | -0.1519                           | 0.7874                           | No          | (-0.4490, 0.1452)   |
| Devarajan et al. (1996)              | 16                 | 0.0431      | 1.6299                                  | 0.0447                            | 1.4274                           | Yes         | (0.0107, 0.0786)    |
| Diamond (1998)                       | 2                  | 0.0393      | 1.0430                                  | 0.0394                            | 1.0397                           | No          | (-0.3286, 0.4074)   |
| Engen and Skinner (1992)             | 6                  | -0.2914     | 1.4635                                  | -0.3843                           | 0.8564                           | Yes         | (-0.7297, -0.0389)  |
| Fölster and Henrekson (1999)         | 7                  | -0.4575     | 0.2512                                  | -0.4693                           | 0.2372                           | Yes         | (-0.5722, -0.3663)  |
| Fölster and Henrekson (2001)         | 8                  | -0.2964     | 0.8113                                  | -0.3579                           | 0.4022                           | Yes         | (-0.4783, -0.2376)  |
| Ghali (2003)                         | 2                  | 0.4323      | 0.0949                                  | 0.4332                            | 0.0947                           | Yes         | (0.0645, 0.8020)    |
| Ghosh and Gregoriou (2008)           | 36                 | 0.2012      | 0.2554                                  | 0.1896                            | 0.2403                           | Yes         | (0.1742, 0.2050)    |
| Grimes (2003)                        | 5                  | -0.4297     | 0.4955                                  | -0.4706                           | 0.4347                           | Yes         | (-0.7247, -0.2165)  |
| Hamdi and Sbia (2013)                | 3                  | 0.0950      | 4.5738                                  | 0.1629                            | 2.7945                           | No          | (-0.9682, 1.2941)   |
| Hansen (1994)                        | 1                  | -0.2133     | n.a.                                    | -0.2133                           | n.a.                             | n.a.        | n.a.                |
| Husnain and Ghani (2010)             | 6                  | -0.1859     | 0.3113                                  | -0.1872                           | 0.3127                           | Yes         | (-0.2486, -0.1258)  |
| Kalaitzidakis and Tzouvelekas (2011) | 1                  | 0.1004      | n.a.                                    | 0.1004                            | n.a.                             | n.a.        | n.a.                |
| Kelly (1997)                         | 4                  | -0.1837     | 0.5736                                  | -0.1877                           | 0.5738                           | Yes         | (-0.3592, -0.0163)  |
| Lee and Lin (1994)                   | 8                  | -0.2528     | 0.2174                                  | -0.2569                           | 0.2145                           | Yes         | (-0.3030, -0.2108)  |

|                                 |            |                |               |                |               |           |                          |
|---------------------------------|------------|----------------|---------------|----------------|---------------|-----------|--------------------------|
| Levine and Renelt (1992)        | 3          | -0.1892        | 0.6099        | -0.1931        | 0.5896        | No        | (-0.4758, 0.0897)        |
| Marlow (1986)                   | 6          | -0.5078        | 0.5296        | -0.5519        | 0.4461        | Yes       | (-0.8102, -0.2935)       |
| Martin and Fardmanesh (1990)    | 12         | 0.0494         | 1.4208        | 0.0361         | 1.9956        | No        | (-0.0097, 0.0820)        |
| Mendoza et al. (1997)           | 3          | -0.0140        | 4.7844        | -0.0059        | 11.7364       | No        | (-0.1789, 0.1670)        |
| Miller and Russek (1997)        | 6          | -0.1513        | 0.7664        | -0.1767        | 0.5151        | Yes       | (-0.2721, -0.0812)       |
| Nketiah-Amponsah (2009)         | 1          | -0.3985        | n.a.          | -0.3985        | n.a.          | n.a.      | n.a.                     |
| Odedokun (1997)                 | 1          | -0.0267        | n.a.          | -0.0267        | n.a.          | n.a.      | n.a.                     |
| Plümpers and Martin (2003)      | 2          | -0.1308        | 0.8283        | -0.1319        | 0.3782        | No        | (-1.1055, 0.8417)        |
| Ram (1986)                      | 8          | -0.2146        | 0.6727        | -0.2074        | 0.6273        | Yes       | (-0.3243, -0.0905)       |
| Romer (1989)                    | 3          | -0.2856        | 0.4083        | -0.3026        | 0.0105        | Yes       | (-0.5869, -0.0183)       |
| Romero-Avila and Strauch (2008) | 3          | -0.1470        | 0.6431        | -0.1534        | 3.7126        | No        | (-0.3934, 0.0858)        |
| Sala-i-Martin (1995)            | 2          | -0.3420        | 0.0105        | -0.3420        | 0.3523        | Yes       | (-0.3743, -0.3096)       |
| Sattar (1993)                   | 9          | 0.0071         | 2.9758        | 0.0047         | 3.7126        | No        | (-0.0087, 0.0181)        |
| Saunders (1985)                 | 2          | -0.6088        | 0.4336        | -0.6847        | 0.3523        | No        | (-2.8519, 1.4825)        |
| Saunders (1988)                 | 12         | -0.4223        | 0.7139        | -0.5150        | 0.5613        | Yes       | (-0.6987, -0.3313)       |
| Scully (1989)                   | 4          | 0.2613         | 0.0994        | 0.2639         | 0.0969        | Yes       | (0.2232, 0.3046)         |
| Stroup and Heckelman (2001)     | 5          | -0.1561        | 1.2757        | -0.1561        | 0.9438        | No        | (-0.3391, 0.0268)        |
| Tanninen (1999)                 | 1          | -0.0360        | n.a.          | -0.0360        | n.a.          | n.a.      | n.a.                     |
| Yan and Gong (2009)             | 8          | 0.0566         | 2.3014        | 0.0594         | 2.7707        | No        | (-0.0782, 0.1970)        |
| <b>Total</b>                    | <b>411</b> | <b>-0.0825</b> | <b>3.1386</b> | <b>-0.0083</b> | <b>9.1092</b> | <b>No</b> | <b>(-0.0238, 0.0071)</b> |

\*Absolute values reported



**Table 1B Means of the partial correlation coefficients per study: Government consumption and growth**

| Study                                  | Reported estimates | Simple mean | *Coefficient of variation (simple mean) | Fixed-effect weighted mean (FEWM) | *Coefficient of variation (FEWM) | Significant | Confidence interval |
|----------------------------------------|--------------------|-------------|-----------------------------------------|-----------------------------------|----------------------------------|-------------|---------------------|
| Afonso and Furceri (2010)              | 4                  | -0.3219     | 0.3772                                  | -0.3023                           | 0.3793                           | Yes         | (-0.4847, -0.1199)  |
| Afonso and Jalles (2014)               | 18                 | -0.1194     | 1.5606                                  | -0.0742                           | 2.4684                           | No          | (-0.1652, 0.0169)   |
| Afonso and Jalles (2013)               | 8                  | -0.1473     | 0.8196                                  | -0.1326                           | 0.5046                           | Yes         | (-0.1886, -0.0767)  |
| Andrés et al. (1996)                   | 2                  | -0.0388     | 0.3889                                  | -0.0388                           | 0.3888                           | No          | (-0.1745, 0.0968)   |
| Angelopoulos and Philippopoulos (2007) | 6                  | -0.3011     | 0.7515                                  | -0.3752                           | 0.5452                           | Yes         | (-0.5899, -0.1605)  |
| Angelopoulos et al. (2008)             | 18                 | -0.1861     | 0.4220                                  | -0.1868                           | 0.3702                           | Yes         | (-0.2211, -0.1524)  |
| Barro and Sala-i-Martin (1995)         | 24                 | -0.3618     | 0.3156                                  | -0.3670                           | 0.2985                           | Yes         | (-0.4133, -0.3208)  |
| Barro (1989)                           | 5                  | -0.4412     | 0.1365                                  | -0.4445                           | 0.1340                           | Yes         | (-0.5185, -0.3705)  |
| Barro (1991)                           | 20                 | -0.4188     | 0.1224                                  | -0.4226                           | 0.1346                           | Yes         | (-0.4492, -0.3960)  |
| Barro (1996)                           | 8                  | -0.2806     | 0.0835                                  | -0.2810                           | 0.0827                           | Yes         | (-0.3004, -0.2615)  |
| Barro (2001)                           | 1                  | -0.6490     | n.a.                                    | -0.6490                           | n.a.                             | n.a.        | n.a.                |
| Belletini and Ceroni (2000)            | 24                 | -0.2187     | 0.6303                                  | -0.2127                           | 0.6031                           | Yes         | (-0.2669, -0.1585)  |
| Bernhard (2001)                        | 1                  | -0.2551     | n.a.                                    | -0.2551                           | n.a.                             | n.a.        | n.a.                |
| Brumm (1997)                           | 1                  | -0.1385     | n.a.                                    | -0.1385                           | n.a.                             | n.a.        | n.a.                |
| Butkiewicz and Yanikkaya (2011)        | 29                 | -0.1199     | 0.8026                                  | -0.1069                           | 0.7060                           | Yes         | (-0.1356, -0.0782)  |
| Castro (2011)                          | 12                 | -0.2969     | 0.5755                                  | -0.3450                           | 0.2771                           | Yes         | (-0.4058, -0.2843)  |
| Commander et al. (1999)                | 9                  | -0.2266     | 0.3393                                  | -0.2173                           | 0.3288                           | Yes         | (-0.2722, -0.1624)  |
| Cooray (2009)                          | 5                  | 0.0165      | 1.3143                                  | 0.0166                            | 1.3136                           | No          | (-0.0105, 0.0436)   |
| Cronovich (1998)                       | 4                  | 0.1728      | 0.9167                                  | 0.1820                            | 0.8977                           | No          | (-0.0780, 0.4420)   |
| De Gregorio (1992)                     | 5                  | -0.1555     | 0.8494                                  | -0.1562                           | 0.8494                           | No          | (-0.3209, 0.0085)   |
| Dowrick (1996)                         | 11                 | -0.0777     | 0.6792                                  | -0.0782                           | 0.7209                           | Yes         | (-0.1160, -0.0403)  |
| Easterly and Rebelo (1993)             | 3                  | -0.0429     | 0.3829                                  | -0.0429                           | 0.3829                           | Yes         | (-0.0837, -0.0021)  |
| Fölster and Henrekson (2001)           | 2                  | -0.3771     | 0.2654                                  | -0.3816                           | 0.2618                           | No          | (-1.2790, 0.5159)   |
| Garrison and Lee (1995)                | 4                  | 0.0067      | 3.4188                                  | 0.0129                            | 1.6873                           | No          | (-0.0217, 0.0475)   |
| Ghura (1995)                           | 6                  | -0.1735     | 0.0860                                  | -0.1737                           | 0.0863                           | Yes         | (-0.1894, -0.1580)  |
| Grier and Tullock (1989)               | 10                 | -0.2161     | 1.2278                                  | -0.2261                           | 0.9932                           | Yes         | (-0.3867, -0.0655)  |
| Grossman (1990)                        | 16                 | 0.0608      | 1.1797                                  | 0.0583                            | 1.2122                           | Yes         | (0.0207, 0.0960)    |
| Guseh (1997)                           | 8                  | -0.0679     | 1.5371                                  | -0.0692                           | 1.5077                           | No          | (-0.1565, 0.0180)   |
| Hansson and Henrekson (1994)           | 6                  | -0.1925     | 0.6625                                  | -0.1967                           | 0.6395                           | Yes         | (-0.3288, -0.0647)  |
| Landau (1983)                          | 14                 | -0.2156     | 0.7223                                  | -0.2222                           | 0.6145                           | Yes         | (-0.3010, -0.1433)  |
| Landau (1986)                          | 12                 | -0.1740     | 0.5860                                  | -0.1025                           | 0.6618                           | Yes         | (-0.1456, -0.0594)  |
| Landau (1997)                          | 8                  | -0.0323     | 1.3497                                  | -0.0311                           | 1.3723                           | No          | (-0.0668, 0.0046)   |
| Lee (1995)                             | 4                  | -0.2972     | 0.1999                                  | -0.3022                           | 0.1994                           | Yes         | (-0.3981, -0.2063)  |
| Levine and Renelt (1992)               | 10                 | -0.2141     | 0.6675                                  | -0.2199                           | 0.6508                           | Yes         | (-0.3224, -0.1176)  |

|                                  |            |                |               |                |               |            |                           |
|----------------------------------|------------|----------------|---------------|----------------|---------------|------------|---------------------------|
| Mo (2007)                        | 10         | -0.4716        | 0.1883        | -0.4806        | 0.1761        | Yes        | (-0.5411, -0.4200)        |
| Murphy et al. (1991)             | 2          | -0.2716        | 0.5109        | -0.3039        | 0.4310        | No         | (-1.4809, 0.8730)         |
| Neycheva (2010)                  | 13         | 0.0273         | 6.0730        | -0.0206        | 5.9556        | No         | (-0.0947, 0.0535)         |
| Romero-Avila and Strauch (2008)  | 5          | -0.1171        | 0.6081        | -0.1177        | 0.6042        | Yes        | (-0.2060, -0.0294)        |
| Roubini and Sala-i-Martin (1992) | 20         | -0.4486        | 0.1186        | -0.4523        | 0.1173        | Yes        | (-0.4771, -0.4274)        |
| Sala-i-Martin (1995)             | 1          | -0.3117        | n.a           | -0.3117        | n.a.          | n.a.       | n.a.                      |
| Saunders (1986)                  | 3          | -0.6019        | 0.3464        | -0.6488        | 0.2878        | Yes        | (-1.1127, -0.1849)        |
| Sheehey (1993)                   | 6          | 0.1206         | 2.4458        | 0.1093         | 2.5298        | No         | (-0.1809, 0.3994)         |
| Tanninen (1999)                  | 3          | -0.0975        | 5.8246        | -0.1855        | 2.8551        | No         | (-1.5008, 1.1299)         |
| Zhang and Casagrande (1998)      | 2          | -0.4290        | 0.0306        | -0.4291        | 0.0306        | Yes        | (-0.5470, -0.3111)        |
| d'Agostino et al. (2010)         | 2          | -0.1162        | 0.6206        | -0.1173        | 0.6149        | No         | (-0.7651, 0.5306)         |
| d'Agostino et al. (2012)         | 3          | -0.1832        | 0.1276        | -0.1833        | 0.1279        | Yes        | (-0.2416, -0.1251)        |
| <b>Total</b>                     | <b>388</b> | <b>-0.2032</b> | <b>0.9813</b> | <b>-0.1204</b> | <b>1.2846</b> | <b>Yes</b> | <b>(-0.1359, -0.1049)</b> |

\*Absolute values reported

**Table 2A Total government expenditures and growth  
PET-FAT-PEESE Results**

| VARIABLES               | PET-FAT               |                        |                     | PEESE                  |
|-------------------------|-----------------------|------------------------|---------------------|------------------------|
|                         | (1)<br>Entire Dataset | (2)<br>Developed       | (3)<br>LDCs         | (4)<br>Developed       |
| Precision ( $\beta_0$ ) | -0.0317<br>(0.0193)   | -0.1311***<br>(0.0459) | -0.0700<br>(0.0467) | -0.1397***<br>(0.0316) |
| Bias ( $\alpha_0$ )     | -0.5963<br>(0.4042)   | 0.0275<br>(0.7804)     | 1.0715<br>(0.7519)  |                        |
| Std. Error              |                       |                        |                     | 4.9584<br>(3.3918)     |
| Observations            | 411                   | 165                    | 139                 | 333                    |
| Number of groups        | 53                    | 28                     | 22                  | 45                     |

Standard errors in parentheses

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

**Table 2B Government consumption and growth  
PET-FAT-PEESE Results**

| VARIABLES               | PET-FAT                |                       |                       | PEESE                  |                        |
|-------------------------|------------------------|-----------------------|-----------------------|------------------------|------------------------|
|                         | (1)<br>Entire Dataset  | (2)<br>Developed      | (3)<br>LDCs           | (4)<br>Entire Dataset  | (5)<br>Developed       |
| Precision ( $\beta_0$ ) | -0.0474***<br>(0.0182) | -0.0862**<br>(0.0403) | -0.0091<br>(0.0320)   | -0.0996***<br>(0.0141) | -0.1397***<br>(0.0260) |
| Bias ( $\alpha_0$ )     | -1.5525***<br>(0.3595) | -1.1544*<br>(0.6206)  | -1.4529**<br>(0.7231) |                        |                        |
| Std. error              |                        |                       |                       | -2.7107<br>(2.0915)    | -2.3687<br>(3.0106)    |
| Observations            | 388                    | 105                   | 70                    | 388                    | 105                    |
| No of studies           | 46                     | 19                    | 14                    | 46                     | 19                     |

Standard errors in parentheses

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

**Table 3A Summary Statistics – Total Expenditure**

| Variables                  | Definition                                                                              | N   | *Mean | #Mean | S.D.  | Min    | Max   |
|----------------------------|-----------------------------------------------------------------------------------------|-----|-------|-------|-------|--------|-------|
| <i>t</i> -value            | t-statistics reported in primary studies                                                | 411 | -0.66 | -0.66 | 2.79  | -12.17 | 6.33  |
| Precision                  | Inverse of standard error of the partial correlation coefficient                        | 411 | 15.07 | 15.07 | 9.99  | 3.16   | 51.03 |
| $SE_{ri}$                  | Standard errors of the partial correlation coefficients                                 | 411 | 0.09  | 0.09  | 0.05  | 0.02   | 0.32  |
| Developed                  | Takes value 1 if the primary study data is from developed countries, otherwise 0        | 411 | 0.40  | 5.09  | 7.39  | 0      | 29.36 |
| LDCs                       | Takes value 1 if the primary study data is from LDCs, otherwise 0                       | 411 | 0.34  | 4.26  | 6.71  | 0      | 33.33 |
| Time Series                | Takes value 1 if Time Series is used by primary study, otherwise 0                      | 411 | 0.07  | 0.55  | 2.01  | 0      | 10.84 |
| Cross-section              | Takes value 1 if cross-section data is used by primary study, 0 if panel is used        | 411 | 0.10  | 0.64  | 2.13  | 0      | 11.04 |
| Panel Data                 | Takes value 1 if panel data is used by primary study, otherwise 0                       | 411 | 0.83  | 13.88 | 11.17 | 0      | 51.03 |
| Control for Endogeneity    | Takes value 1 if primary study controls for endogeneity, otherwise 0                    | 411 | 0.17  | 2.22  | 5.35  | 0      | 22.41 |
| Endogenous Growth Model    | Takes value 1 if the model is based on endogenous growth model, otherwise 0.            | 411 | 0.42  | 0.06  | 0.75  | 0      | 10.11 |
| Data Average (>=5)         | Takes value 1 if data averaging period is >=5 years otherwise 0                         | 411 | 0.47  | 6.05  | 7.03  | 0      | 23.36 |
| Data Average*Panel Data    | Takes value 1 if study used panel data and averaging period is >=5 years otherwise 0    | 411 | 0.54  | 6.94  | 7.23  | 0      | 27.50 |
| Data Average*Cross Section | Takes value 1 if study used cross section and averaging period is >=5 years otherwise 0 | 411 | 0.09  | 0.64  | 2.13  | 0      | 11.04 |
| Data Period (1960+)        | Takes value 1 if data year >= 1960, otherwise 0                                         | 411 | 0.94  | 12.97 | 11.51 | 0      | 51.03 |
| Data Period (1970+)        | Takes value 1 if data year >= 1970, otherwise 0                                         | 411 | 0.88  | 8.38  | 12.48 | 0      | 51.03 |
| Data Period (1980+)        | Takes value 1 if data year >= 1980, otherwise 0                                         | 411 | 0.82  | 11.19 | 7.43  | 0      | 44.44 |
| Data Period (1990+)        | Takes value 1 if data year >= 1990, otherwise 0                                         | 411 | 0.49  | 13.29 | 10.98 | 0      | 51.03 |
| Data Period (2000+)        | Takes value 1 if data year >= 2000, otherwise 0                                         | 411 | 0.09  | 14.19 | 10.45 | 0      | 51.03 |
| Initial GDP                | Takes value 1 if the primary study control for initial per capita GDP, otherwise 0      | 411 | 0.44  | 5.49  | 7.02  | 0      | 26.26 |
| Population                 | Takes value 1 if the primary study control for population, otherwise 0                  | 411 | 0.17  | 1.84  | 4.10  | 0      | 18.71 |
| Investment                 | Takes value 1 if the primary study control for investment, otherwise 0                  | 411 | 0.46  | 5.98  | 8.04  | 0      | 44.44 |
| Tax                        | Takes value 1 if the primary study control for taxes, otherwise 0                       | 411 | 0.38  | 3.99  | 7.01  | 0      | 27.50 |
| Journal Rank               | Takes value 1 if the primary study is published in high-ranked journal, otherwise 0     | 411 | 0.42  | 6.79  | 7.32  | 0      | 29.69 |
| Journal                    | Takes value 1 if the primary study is published in a journal, otherwise 0               | 411 | 0.81  | 10.48 | 7.71  | 0      | 44.44 |
| Publication Year (1990+)   | Takes value 1 if publication year >=1990, otherwise 0                                   | 411 | 0.82  | 3.05  | 7.29  | 0      | 44.44 |
| Publication Year (2000+)   | Takes value 1 if publication year >=2000, otherwise 0                                   | 411 | 0.59  | 4.67  | 6.79  | 0      | 29.69 |
| Publication Year (2010+)   | Takes value 1 if publication year >=2010, otherwise 0                                   | 411 | 0.34  | 6.79  | 12.09 | 0      | 51.03 |

Notes: \*un-weighted mean, #weighted mean, and weighted variables are divided by  $SE_{ri}$

**Table 3B Summary Statistics – Government Consumption**

| Variables                  | Definition                                                                              | N   | *Mean | #Mean | S.D.  | Min    | Max   |
|----------------------------|-----------------------------------------------------------------------------------------|-----|-------|-------|-------|--------|-------|
| <i>t</i> -value            | t-statistics reported in primary studies                                                | 388 | -2.24 | -2.24 | 2.13  | -10.32 | 3.53  |
| Precision                  | Inverse of standard error of the partial correlation coefficient                        | 388 | 13.49 | 13.49 | 8.17  | 3.25   | 43.49 |
| $SE_{ri}$                  | Standard errors of the partial correlation coefficients                                 | 388 | 0.10  | 0.10  | 0.05  | 0.02   | 0.31  |
| Developed                  | Takes value 1 if the primary study data is from developed countries, otherwise 0        | 388 | 0.27  | 3.33  | 6.34  | 0      | 26.94 |
| LDCs                       | Takes value 1 if the primary study data is from LDCs, otherwise 0                       | 388 | 0.18  | 3.54  | 8.68  | 0      | 39.73 |
| Time Series                | Takes value 1 if Time Series is used by primary study, otherwise 0                      | 388 | 0.02  | 0.08  | 0.67  | 0      | 6.48  |
| Cross-section              | Takes value 1 if cross-section data is used by primary study, 0 if panel is used        | 388 | 0.31  | 2.91  | 4.87  | 0      | 24.89 |
| Panel Data                 | Takes value 1 if panel data is used by primary study, otherwise 0                       | 388 | 0.67  | 10.49 | 10.30 | 0      | 43.49 |
| Control for Endogeneity    | Takes value 1 if primary study controls for endogeneity, otherwise 0                    | 388 | 0.26  | 1.99  | 5.03  | 0      | 26.45 |
| Endogenous Growth Model    | Takes value 1 if the model is based on endogenous growth model, otherwise 0.            | 388 | 0.25  | 0.26  | 1.59  | 0      | 11.01 |
| Data Average (>=5)         | Takes value 1 if data averaging period is >=5 years otherwise 0                         | 388 | 0.31  | 7.28  | 7.47  | 0      | 27.16 |
| Data Average*Panel Data    | Takes value 1 if study used panel data and averaging period is >=5 years otherwise 0    | 388 | 0.40  | 5.19  | 7.57  | 0      | 27.16 |
| Data Average*Cross Section | Takes value 1 if study used cross section and averaging period is >=5 years otherwise 0 | 388 | 0.21  | 2.09  | 4.49  | 0      | 24.89 |
| Data Period (1960+)        | Takes value 1 if data year >= 1960, otherwise 0                                         | 388 | 0.96  | 7.49  | 9.15  | 0      | 43.49 |
| Data Period (1970+)        | Takes value 1 if data year >= 1970, otherwise 0                                         | 388 | 0.86  | 9.12  | 8.91  | 0      | 42.60 |
| Data Period (1980+)        | Takes value 1 if data year >= 1980, otherwise 0                                         | 388 | 0.71  | 12.80 | 8.52  | 0      | 12.51 |
| Data Period (1990+)        | Takes value 1 if data year >= 1990, otherwise 0                                         | 388 | 0.51  | 12.11 | 9.02  | 0      | 25.26 |
| Data Period (2000+)        | Takes value 1 if data year >= 2000, otherwise 0                                         | 388 | 0.23  | 0.14  | 1.23  | 0      | 11.01 |
| Initial GDP                | Takes value 1 if the primary study control for initial per capita GDP, otherwise 0      | 388 | 0.67  | 8.48  | 8.32  | 0      | 43.49 |
| Population                 | Takes value 1 if the primary study control for population, otherwise 0                  | 388 | 0.39  | 5.09  | 7.59  | 0      | 26.94 |
| Investment                 | Takes value 1 if the primary study control for investment, otherwise 0                  | 388 | 0.60  | 7.25  | 7.95  | 0      | 42.60 |
| Tax                        | Takes value 1 if the primary study control for taxes, otherwise 0                       | 388 | 0.34  | 2.49  | 7.62  | 0      | 43.49 |
| Journal Rank               | Takes value 1 if the primary study is published in high-ranked journal, otherwise 0     | 388 | 0.23  | 8.68  | 9.88  | 0      | 43.49 |
| Journal                    | Takes value 1 if the primary study is published in a journal, otherwise 0               | 388 | 0.87  | 11.96 | 9.27  | 0      | 43.49 |
| Publication Year (1990+)   | Takes value 1 if publication year >=1990, otherwise 0                                   | 388 | 0.83  | 6.89  | 8.38  | 0      | 34.36 |
| Publication Year (2000+)   | Takes value 1 if publication year >=2000, otherwise 0                                   | 388 | 0.51  | 11.79 | 9.36  | 0      | 43.49 |
| Publication Year (2010+)   | Takes value 1 if publication year >=2010, otherwise 0                                   | 388 | 0.23  | 3.16  | 6.56  | 0      | 27.16 |

Notes: \*un-weighted mean, #weighted mean, and weighted variables are divided by  $SE_{ri}$

**Table 4A – MRA (Total Government Expenditure and Growth)**

| VARIABLES                                     | (1)<br>General Model   | (2)<br>General-to-Specific Model |
|-----------------------------------------------|------------------------|----------------------------------|
| Precision                                     | -0.0641<br>(0.1510)    | 0.0397<br>(0.0880)               |
| <i>Theoretical and econometric dimensions</i> |                        |                                  |
| Control for Endogeneity                       | 0.0669***<br>(0.0203)  | 0.0649***<br>(0.0203)            |
| Time Series                                   | -0.2058<br>(0.1298)    |                                  |
| Cross Section                                 | -0.1145<br>(0.1224)    |                                  |
| Endogenous Growth Model                       | 0.3526<br>(0.2751)     |                                  |
| Data Average (=>5 years)                      | 0.0028<br>(0.0829)     |                                  |
| Data Average*Panel Data                       | -0.0338<br>(0.0799)    |                                  |
| Population                                    | -0.1892***<br>(0.0707) | -0.1739***<br>(0.0605)           |
| Initial GDP                                   | -0.1837***<br>(0.0422) | -0.1733***<br>(0.0375)           |
| Investment                                    | -0.0677*<br>(0.0374)   | -0.0574*<br>(0.0347)             |
| <i>Data Characteristics</i>                   |                        |                                  |
| Data Period (1960+)                           | 0.0254<br>(0.0317)     |                                  |
| Data Period (1970+)                           | 0.0234<br>(0.0508)     |                                  |
| Data Period (1980+)                           | -0.0201<br>(0.0687)    |                                  |
| Data Period (1990+)                           | 0.0091<br>(0.0618)     |                                  |
| Data Period (2000+)                           | -0.1065<br>(0.0946)    | -0.1232*<br>(0.0698)             |
| Developed                                     | -0.0304*<br>(0.0171)   | -0.0312**<br>(0.0159)            |
| <i>Publication Characteristics</i>            |                        |                                  |
| Journal Rank                                  | 0.0225<br>(0.0484)     |                                  |
| Journal                                       | 0.2243***<br>(0.0670)  | 0.1894***<br>(0.0483)            |
| Publication Year (1990+)                      | -0.0632<br>(0.0824)    | -0.0921*<br>(0.0540)             |
| Publication Year (2000+)                      | -0.0634<br>(0.0721)    | -0.0839*<br>(0.0430)             |
| Publication Year (2010+)                      | 0.1697**<br>(0.0770)   | 0.1682***<br>(0.0451)            |
| Constant                                      | 1.1650**<br>(0.5104)   | 0.7136<br>(0.4440)               |
| Observations                                  | 411                    | 411                              |
| Number of groups                              | 53                     | 53                               |

Standard errors in parentheses.

\*\*\* p&lt;0.01, \*\* p&lt;0.05, \* p&lt;0.1

**Table 4B – MRA (Government Consumption and Growth)**

| VARIABLES                                     | (1)<br>General Model   | (2)<br>General-to-Specific Model |
|-----------------------------------------------|------------------------|----------------------------------|
| Precision                                     | 0.0737<br>(0.1509)     | 0.1776<br>(0.1256)               |
| <i>Theoretical and econometric dimensions</i> |                        |                                  |
| Control for Endogeneity                       | 0.0256<br>(0.0243)     |                                  |
| Time Series                                   | 0.0140<br>(0.1701)     |                                  |
| Cross Section                                 | -0.2740***<br>(0.0627) | -0.2328***<br>(0.0579)           |
| Endogenous Growth Model                       | -0.1376**<br>(0.0608)  | -0.1226**<br>(0.0603)            |
| Data Average (=>5)                            | -0.0232<br>(0.0304)    |                                  |
| Data Average*Cross Section                    | 0.1320*<br>(0.0733)    | 0.1029*<br>(0.0634)              |
| Population                                    | -0.0765***<br>(0.0268) | -0.0713***<br>(0.0243)           |
| Initial GDP                                   | 0.0256<br>(0.0234)     |                                  |
| Tax                                           | -0.0373<br>(0.0228)    |                                  |
| Investment                                    | 0.0937***<br>(0.0200)  | 0.0908***<br>(0.0194)            |
| <i>Data Characteristics</i>                   |                        |                                  |
| Data Period (1960+)                           | -0.0687<br>(0.0468)    | -0.0837**<br>(0.0425)            |
| Data Period (1970+)                           | -0.0985**<br>(0.0462)  | -0.1216***<br>(0.0402)           |
| Data Period (1980+)                           | -0.1214**<br>(0.0530)  | -0.1487***<br>(0.0475)           |
| Data Period (1990+)                           | -0.2309***<br>(0.0686) | -0.2760***<br>(0.0501)           |
| Developed                                     | -0.0138<br>(0.0180)    |                                  |
| <i>Publication Characteristics</i>            |                        |                                  |
| Journal Rank                                  | -0.0761***<br>(0.0287) | -0.0866***<br>(0.0265)           |
| Journal                                       | 0.2115***<br>(0.0440)  | 0.2040***<br>(0.0424)            |
| Publication Year (1990+)                      | 0.0531**<br>(0.0241)   | 0.0408*<br>(0.0231)              |
| Publication Year (2000+)                      | 0.1915***<br>(0.0389)  | 0.1921***<br>(0.0360)            |
| Publication Year (2010+)                      | -0.1669***<br>(0.0359) | -0.1822***<br>(0.0312)           |
| Constant                                      | -1.1041***<br>(0.2932) | -1.0963***<br>(0.2747)           |
| Observations                                  | 388                    | 388                              |
| Number of groups                              | 46                     | 46                               |

Standard errors in parentheses. \*\*\* p&lt;0.01, \*\* p&lt;0.05, \* p&lt;0.1

## Appendix – Overview of Methods

### 1. Partial Correlation Coefficients (PCCs)

PCCs measure the association between government expenditure and per-capita GDP growth. Given that they are independent of the metrics used in measuring both independent and dependent variables, they allow for the comparability of studies and reported effect-size estimates. They are mostly used in meta-analysis (see e.g. Alptekin and Levine, 2012; Ugur; 2014; Benos and Zotou, 2014).

We use equations (A1) and (A2) to calculate a PCC and standard error, respectively, for each relevant effect-size estimate reported by primary studies.

$$r_i = \frac{t_i}{\sqrt{t_i^2 + df_i}} \quad (A1)$$

and

$$SE_{ri} = \sqrt{\frac{1 - r_i^2}{df_i}} \quad (A2)$$

$r_i$  and  $SE_{ri}$  represent PCC and its associated standard errors, respectively.  $SE_{ri}$  represents variations due to sampling error and its inverse is used as weight in the calculation of study-by-study fixed-effect weighted averages.  $t_i$  and  $df_i$  represent  $t$ -value and degrees of freedom, respectively, associated with estimates reported in primary studies.

### 2. Fixed Effect Weighted Means

We calculate FEEs using (A3) below.

$$\bar{X}_{FEE} = \frac{\sum r_i \left( \frac{1}{SE_{ri}^2} \right)}{\sum \frac{1}{SE_{ri}^2}} \quad (A3)$$

$\bar{X}_{FEE}$  is the fixed effect weighted average and all other variables remain as explained before. FEEs account for within-study variations by assigning higher weights to more precise estimates, and lower weights to less precise estimates.

### 3. Bivariate meta-regressions

To estimate ‘genuine effect’ beyond publication selection bias, we draw on meta-regression analysis (MRA) models proposed and developed by Stanley (2008) and Stanley and Doucouliagos (2012,



2014). The underpinning theoretical framework is that of Egger et al. (1997), who postulate that researchers with small samples and large standard errors would search intensely across model specifications, econometric techniques and data measures to find sufficiently large (hence statistically significant) effect-size estimates. Hence:

$$e_i = \beta + \alpha SE_i + u_i \quad (A4)$$

Here,  $e_i$  is the effect-size reported in primary studies and  $SE_i$  is the associated standard error. Rejecting the null hypothesis of  $\alpha = 0$  indicates the presence of publication bias. This is also known as the funnel-asymmetry test (FAT), which evaluates the asymmetry of the funnel graphs that chart the effect-size estimates against their precisions. Testing for  $\beta = 0$  is known as precision-effect test (PET), and allows for establishing whether genuine effect exists beyond selection bias.

However, estimating (A4) poses several issues. First, the model is heteroskedastic: effect-size estimates have widely different standard errors (hence variances), violating the assumption of independently and identically distributed (i.i.d.) error term ( $u_i$ ). To address this issue, Stanley (2008) and Stanley and Doucouliagos (2012) propose a weighted least squares (WLS) version, obtained by dividing both sides of (A4) with precision ( $1/SE_i$ ). Secondly, the reported estimates may be subject to study-specific random effects that differ between studies. To address this issue, it is necessary to use a hierarchical linear model (HLM) that allows for nesting the individual estimates within studies. The appropriateness of the HLM should be verified by comparing it with the standard OLS estimation, using a likelihood ratio (LR) test. The HLM model can be stated as follows:

$$t_{ij} = \alpha_0 + \beta_0 \left( \frac{1}{SE_{rij}} \right) + v_j + \varepsilon_{ij} \quad (A5)$$

where  $t_{ij}$  is the  $t$ -value associated with effect-size estimate  $i$  (i.e., the partial correlation coefficient calculated using A1) of study  $j$ ;  $SE_{rij}$  is the corresponding standard calculated in accordance with (A2);  $v_j$  is the study-level random effect; and  $\varepsilon_{ij}$  is the multivariate-normal error term with mean zero. The random effects ( $v_j$ ) are not estimated directly, but their variance (or standard error) is. We conclude in favour of publication selection bias if  $\alpha_0$  is statistically significant at conventional levels. In the presence of publication bias,  $\alpha_0$  determines the magnitude and the direction of bias. Similarly, we conclude in favour of genuine effect beyond selection bias if  $\beta_0$  is statistically significant at conventional levels.

The third issue is that Egger et al. (1997) assume a linear relationship between primary-study estimates and their standard errors. However, Moreno et al (2009) and Stanley and Doucouliagos (2014) provide simulation evidence indicating that a quadratic specification is superior if 'genuine

effect' exists beyond selection bias – i.e., if the PET in (A5) rejects the null hypothesis of zero effect. Then, the correct specification is referred to as precision-effect test corrected for standard errors (PEESE) and can be stated as follows:

$$t_{ij} = \beta_0 \left( \frac{1}{SE_{rij}} \right) + \alpha_0 (SE_{rij}) + v_j + \varepsilon_{ij} \quad (A6)$$

Given that study-level random effects may be observed at the intercept or slope levels or both, we establish which type of HML is appropriate using LR tests, where the null hypothesis is that the preferred specification is nested within the comparator specification. Therefore, we estimate HLMS with random-intercepts only and HLMS with random intercepts and random slopes; and test whether the latter are nested within the former.

#### 4. Multivariate Meta-Regression Model (MRM)

To address the issues of heterogeneity, we estimate a multivariate hierarchical meta-regression model specified in (A7) below.

$$t_{ji} = \alpha_0 + \beta_0 \left( \frac{1}{SE_{jri}} \right) + \sum \beta_k \frac{(Z_{ki})}{SE_{jri}} + v_j + \varepsilon_{ij} \quad (A7)$$

Here,  $t_{ji}$  is the  $i$ th  $t$ -value from the  $j$ th study, while  $Z_{ki}$  is a  $k \times 1$  vector of moderator variables that capture the observable sources of heterogeneity in the government size-growth evidence base.

To minimise the risk of multicollinearity and over-fitting, we estimate (A7) through a general-to-specific estimation routine, whereby we omit the most insignificant variables (variables associated with the largest p-values) one at a time until all remaining covariates are statistically significant. We present the findings from the specific and general models side by side to: (a) establish the extent of congruence between the significant moderating factors; and (ii) identify the range of moderating variables that do not affect the variation in the evidence base.